Emad Mahmodi

Software Engineer · Data Scientist · Security & Reverse-Engineering Researcher

Machine Learning & Data Science | Algorithms & HPC | AI for Security

Profile

Software engineer and applied-ML researcher with 10+ years spanning software engineering, data science, high-performance computing, and security. Three peer-reviewed machine-learning publications (anomaly detection, recommender systems; Google Scholar h-index 3). Strength in turning hard, structured data — binaries, graphs, network traffic — into working ML and systems, backed by deep automotive reverse-engineering expertise.

Research Interests

- **Machine Learning & Data Science:** anomaly detection, concept drift, cross-domain generalization, recommender systems.
- **Algorithms & High-Performance Computing:** parallel and GPU (CUDA/OpenMP) algorithms on irregular, power-law data.
- **AI for Cybersecurity:** LLMs and RAG for automated binary analysis and firmware documentation.
- **Binary Reverse Engineering:** automotive ECUs (ISA reconstruction, Seed/Key bypass, checksum extraction).

Education

M.Sc. in Computer Science (Software Engineering)

Ferdowsi University of Mashhad, Iran | 2014 – 2017

GPA: 16.71/20 (final year) | Thesis: Adaptive rule-based phishing detection against URL obfuscation (18/20). Supervisors: Dr. H. Sadoghi Yazdi, Dr. A. Ghaemi Bafghi.

B.Sc. in Computer Engineering (Software)

Shomal University of Amol, Iran | 2008 – 2013

GPA: 17.14/20 (final year) | Final project: Integrated online sales management system (20/20).

Publications

- H. Koohi, Z. Kobti, T. Farzi, **E. Mahmodi** (2024). "UDIS: Enhancing Collaborative Filtering with Fusion of Dimensionality Reduction and Semantic Similarity." *Electronics*, 13(20), 4073. MDPI.
- **E. Mahmodi**, H. Sadoghi Yazdi, A. Ghaemi Bafghi (2020). "A drift-aware adaptive method based on minimum uncertainty for anomaly detection in data streams with skewed distribution." *Expert Systems with Applications*, Elsevier.
- R. Barzegar Nozari, H. Koohi, **E. Mahmodi** (2020). "A Novel Trust Computation Method Based on User Ratings to Improve the Recommendation." *International Journal of Engineering (IJE)*.
- **E. Mahmodi**, et al. "Intelligent detection scheme using adaptive feature reduction for phishing attack." *Future Generation Computer Systems*, Elsevier (Under Review).
- K. Mahmoudi, **E. Mahmodi**. "Distributed deep learning approach for sentiment analysis using BiLSTM & GloVe Embedding." *International Journal of Engineering* (Submitted).

Professional Experience

Senior Software Developer & Lead Reverse Engineer | Negar-Khodro

2022 – Present

- Architected ECUProg v2, an automotive programming platform (C++17/Qt 6): 49% lower memory footprint, 62% less JTAG overhead.

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- Led reverse engineering of TriCore (Infineon) and PowerPC (VLE) firmware; reconstructed UDS/KWP2000 diagnostic logic.
- Designed 'Negaremap' (Persian remapping platform) and prototyped an LLM/RAG pipeline plus custom Ghidra/IDA Pro plugins.

Lecturer & M.Sc. Thesis Advisor | Shomal University, Amol
2017 – 2024 (part-time)

- Taught the graduate course 'Principles of Data Mining' (2019–2020).
- Advised two M.Sc. theses on recommender systems (group recommendation; collaborative filtering with dimensionality reduction & semantic methods) — both students co-authored publications with me.

Senior System Programmer | NIUniverse
2020 – 2022

- Binary security auditing and vulnerability research: bypassed anti-debugging and V-table obfuscation; built high-speed PC-to-hardware drivers.

R&D Programmer / Data Scientist | Ferdowsi University — DCSL & Pattern Recognition Lab
2015 – 2020

- Built data-mining, anomaly-detection, and recommender-system pipelines (C++, Python, SQL) underpinning the ESWA (2020) and IJE (2020) publications.

Selected Open-Source Projects

- **GPU-Accelerated Graph Analytics** (github.com/Emad-Mahmodi/gpu-graph-study): reproducible study of BFS/PageRank/WCC/Louvain on power-law security graphs — C++17, OpenMP, CUDA.
- **Cross-Domain IDS Generalization** (github.com/Emad-Mahmodi/adaptive-ids-generalization): PyTorch drift-aware ensemble on NSL-KDD; quantified a 21-point in-domain→cross-domain F1 gap.
- **Context-Augmented RAG for Binary Analysis** (github.com/Emad-Mahmodi/binary-rag-copilot): lightweight function-name recovery; independent take on the ReCopilot idea, no model training.
- **PISAD ECU Simulator (PowerPC)** & **Bajaj MSE2 ECU security audit**: firmware emulation and Seed/Key handshake analysis.

Technical Skills

- **Languages:** C/C++17/20, Python, Java, MATLAB, Assembly (TriCore, PowerPC, x86).
- **ML & Data Science:** PyTorch, scikit-learn, anomaly detection, concept drift, ensemble methods, LLM fine-tuning / RAG.
- **Algorithms & HPC:** CUDA, OpenMP, SIMD (AVX2/SSE), parallel graph algorithms, performance engineering.
- **Data Engineering:** SQL, data pipelines & ETL, large-scale structured data.
- **Reverse Engineering & Security:** Ghidra, IDA Pro, Radare2, binary diffing, UDS / CAN-Bus / OBD-II.
- **Software Engineering:** Qt/QML, Git, CMake, CI/CD, Linux.

Certificates & Service

- Reviewer, 10th International Conference on Computer and Knowledge Engineering (ICCKE), 2020.
- Machine Learning (Stanford) · IBM Data Science · LPIC-1/2 Linux Professional.
- Languages: Persian (native), English (advanced; TOEFL candidate 2026).

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